使用Cisco Nexus 9336C-FX2叢集交換器移轉至雙節點交換叢集

ONTAP Systems Switches

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap-systems-switches/switch-cisco-9336c-fx2/migrate-to-2n-switched-overview.html on May 03, 2022. Always check docs.netapp.com for the latest.

目錄

使用Cisco Nexus 9336C-FX2叢集交換器移轉至雙節點交換叢集 1
使用Cisco Nexus 9336C-FX2叢集交換器移轉至雙節點交換叢集 1
如何使用Cisco Nexus 9336C-FX2叢集交換器移轉至交換式NetApp叢集環境 1

使用Cisco Nexus 9336C-FX2叢集交換器移轉至雙節點交換叢集

使用Cisco Nexus 9336C-FX2叢集交換器移轉至雙節點交換叢集

將雙節點無交換式叢集（不中斷營運）移轉至採用Cisco Nexus 9336C-FX2叢集交換器的叢集時、您必須瞭解某些組態資訊、連接埠連線和纜線需求。您使用的程序取決於每個控制器上是否有兩個專屬的叢集網路連接埠、或每個控制器上是否有一個叢集連接埠。記錄的程序適用於所有使用光纖或雙軸纜線連接埠的節點、但如果節點使用內建10Gb Base-T RJ45連接埠作為叢集網路連接埠、則此交換器不支援此程序。

大多數系統在每個控制器上都需要兩個專用的叢集網路連接埠。請參閱 "Cisco乙太網路交換器"

如何使用Cisco Nexus 9336C-FX2叢集交換器移轉至交換式NetApp叢集環境

如果您現有的雙節點無交換式叢集環境、可以使用Cisco Nexus 9336C-FX2交換器移轉至雙節點交換式叢集環境、以便擴充至叢集中的兩個節點以外。

雙節點無交換器組態：

- 雙節點無交換器組態必須正確設定並正常運作。
- 節點必須執行ONTAP 的是更新版本的版本。
- 所有叢集連接埠都必須處於「up」（開機）狀態。
- 所有叢集邏輯介面（LIF）都必須處於「up」（開機）狀態、並位於其主連接埠上。

Cisco Nexus 9336C-FX2交換器組態：

- 兩台交換器都必須具備管理網路連線能力。
- 叢集交換器必須有主控台存取權。
- Nexus 9336C-FX2節點對節點交換器和交換器對交換器連線必須使用雙軸纜線或光纖纜線。
  - "交換器Hardware Universe" 包含纜線的詳細資訊。

  "交換器Hardware Universe"
- 交換器間連結（ISL）纜線必須連接至9336C-FX2交換器的連接埠1/35和1/36。
- 必須完成9336C-FX2交換器的初始自訂。因此：
  - 9336C-FX2交換器正在執行最新版本的軟體
  - 參考組態檔（RCT）已套用至交換器任何站台自訂、例如必須在新交換器上設定任何站台、例如：SMTP、SNMP和SSH。

本程序中的範例使用下列叢集交換器和節點命名法：

- 9336C-FX2交換器的名稱為CS1和CS2。
- 叢集SVM的名稱為node1和node2。
- lifs的名稱分別是節點1上的node1_clus1和node1_clus2、以及節點2上的node2_clus1和node2_clus2。
- 「cluster1：*：>」提示會指出叢集的名稱。
- 此程序中使用的叢集連接埠為e0a和e0b。
  - "Hardware Universe" 包含平台實際叢集連接埠的最新資訊。

步驟

1. 如果AutoSupport 此叢集啟用了「支援功能」、請叫用AutoSupport 「系統節點AutoSupport 支援功能叫用節點*-type all -most MAn=xh」訊息來禁止自動建立案例

   其中x是維護時段的持續時間（以小時為單位）。

   > 此資訊可通知技術支援人員執行此維護工作、以便在維護期間抑制自動建立案例。AutoSupport

2. 請將權限等級變更為進階、並在系統提示您繼續時輸入「y」：「et -priv榮幸 進階」

   出現進階提示（「*>」）。

3. 在新的叢集交換器CS1和CS2上停用所有面向節點的連接埠（非ISL連接埠）。

   您不得停用ISL連接埠。

   以下範例顯示、交換器CS1上的節點對介連接埠1至34已停用：

   ```
   cs1# config
   Enter configuration commands, one per line. End with CNTL/Z.
   cs1(config)# interface e/1-34
   cs1(config-if-range)# shutdown
   ```

4. 驗證ISL和ISL上兩個9336C-FX2交換器CS1和CS2之間的實體連接埠是否在連接埠1/35和1/36上：「How port-channel Summary（顯示連接埠通道摘要）」

   下列範例顯示交換器CS1上的ISL連接埠已開啟：

```
cs1# show port-channel summary

Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
--------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------------
--------
1     Po1(SU)     Eth      LACP      Eth1/35(P)   Eth1/36(P)
```

以下範例顯示交換器CS2上的ISL連接埠已開啟：

```
(cs2)# show port-channel summary

Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
--------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------------
--------
1     Po1(SU)     Eth      LACP      Eth1/35(P)   Eth1/36(P)
```

5. 顯示鄰近裝置的清單：「How cup neighbor」

   此命令提供有關連線至系統之裝置的資訊。

   下列範例列出交換器CS1上的鄰近裝置：

```
cs1# show cdp neighbors

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
ID
cs2                Eth1/35        175    R S I s     N9K-C9336C
Eth1/35
cs2                Eth1/36        175    R S I s     N9K-C9336C
Eth1/36

Total entries displayed: 2
```

下列範例列出交換器CS2上的鄰近裝置：

```
cs2# show cdp neighbors

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
ID
cs1                Eth1/35        177    R S I s     N9K-C9336C
Eth1/35
cs1           )    Eth1/36        177    R S I s     N9K-C9336C
Eth1/36

Total entries displayed: 2
```

6. 驗證所有叢集連接埠是否正常運作：「network port show -IPSpace Cluster（網路連接埠顯示-IPSpace叢集）」

   每個連接埠應顯示「Link」（連結）、「Health Status」（健全狀況狀態）則顯示為「Healthy」（健全狀況）。

```
cluster1::*> network port show -ipspace Cluster

Node: node1

                                                   Speed(Mbps) Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
--------- ------------ ---------------- ---- ---- ----------- --------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
e0b       Cluster      Cluster          up   9000  auto/10000 healthy

Node: node2

                                                   Speed(Mbps) Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
--------- ------------ ---------------- ---- ---- ----------- --------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
e0b       Cluster      Cluster          up   9000  auto/10000 healthy

4 entries were displayed.
```

7. 驗證所有叢集生命體是否正常運作：「network interface show -vserver cluster」（網路介面show -vserver 叢集）

   每個叢集LIF都應該顯示為「IS Home」、並具有「狀態管理/作業」的「啟動/啟動」

```
cluster1::*> network interface show -vserver Cluster

            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- -----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1         e0a
true
            node1_clus2  up/up    169.254.49.125/16  node1         e0b
true
            node2_clus1  up/up    169.254.47.194/16  node2         e0a
true
            node2_clus2  up/up    169.254.19.183/16  node2         e0b
true
4 entries were displayed.
```

8. 驗證是否已在所有叢集lifs上啟用自動還原：「network interface show -vserver cluster -Fields auto-Revert」

```
cluster1::*> network interface show -vserver Cluster -fields auto-revert

          Logical
Vserver   Interface     Auto-revert
--------- ------------- ------------
Cluster
          node1_clus1   true
          node1_clus2   true
          node2_clus1   true
          node2_clus2   true

4 entries were displayed.
```

9. 從節點1上的叢集連接埠e0a拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e0a連接至叢集交換器CS1上的連接埠1。

   - "交換器Hardware Universe" 包含纜線的詳細資訊。

   "交換器Hardware Universe"

10. 從節點2上的叢集連接埠e0a拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e0a連接至叢集交換器CS1上的連接埠2。
11. 在叢集交換器CS1上啟用所有面向節點的連接埠。

    下列範例顯示交換器CS1上已啟用連接埠1/1至1/34：

```
cs1# config
Enter configuration commands, one per line. End with CNTL/Z.
cs1(config)# interface e1/1-34
cs1(config-if-range)# no shutdown
```

12. 驗證所有叢集生命體是否都正常運作、並顯示為「is Home」（原為主目錄）：「network interface show -vserver Cluster」（網路介面show -vserver叢集）

    以下範例顯示、節點1和節點2上的所有lifs都已啟動、而且「是主目錄」結果為真：

```
cluster1::*> network interface show -vserver Cluster

         Logical      Status     Network            Current     Current
Is
Vserver  Interface    Admin/Oper Address/Mask       Node        Port
Home
-------- ------------ ---------- ------------------ ----------- -------
----
Cluster
         node1_clus1  up/up      169.254.209.69/16  node1       e0a
true
         node1_clus2  up/up      169.254.49.125/16  node1       e0b
true
         node2_clus1  up/up      169.254.47.194/16  node2       e0a
true
         node2_clus2  up/up      169.254.19.183/16  node2       e0b
true

4 entries were displayed.
```

13. 顯示叢集中節點狀態的相關資訊：「叢集顯示」

    下列範例顯示叢集中節點的健全狀況和資格資訊：

```
cluster1::*> cluster show

Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------  ------------
node1                true    true          false
node2                true    true          false

2 entries were displayed.
```

14. 從節點1上的叢集連接埠e0b拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e0b連接至叢集交換器CS2上的連接埠1。
15. 從節點2上的叢集連接埠e0b拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e0b連接至叢集交換器CS2上的連接埠2。
16. 在叢集交換器CS2上啟用所有面向節點的連接埠。

    下列範例顯示交換器CS2上已啟用連接埠1/1至1/34：

```
cs2# config
Enter configuration commands, one per line. End with CNTL/Z.
cs2(config)# interface e1/1-34
cs2(config-if-range)# no shutdown
```

17. 驗證所有叢集連接埠是否正常運作：「network port show -IPSpace Cluster（網路連接埠顯示-IPSpace叢集）」

    以下範例顯示節點1和節點2上的所有叢集連接埠都已啟動：

```
cluster1::*> network port show -ipspace Cluster

Node: node1

Ignore
                                                   Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

Node: node2

Ignore
                                                   Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
e0a       Cluster      Cluster          up   9000  auto/10000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/10000 healthy
false

4 entries were displayed.
```

18. 驗證所有介面是否都顯示為「is Home」（原為主介面）：「network interface show -vserver Cluster（網路介面show -vserver叢集）」

這可能需要幾分鐘的時間才能完成。

以下範例顯示所有lifs都在node1和node2上、而且「is Home」結果為真：

```
cluster1::*> network interface show -vserver Cluster

          Logical      Status     Network            Current    Current
Is
Vserver   Interface    Admin/Oper Address/Mask       Node       Port
Home
--------- ------------ ---------- ------------------ ---------- -------
----
Cluster
          node1_clus1  up/up      169.254.209.69/16  node1      e0a
true
          node1_clus2  up/up      169.254.49.125/16  node1      e0b
true
          node2_clus1  up/up      169.254.47.194/16  node2      e0a
true
          node2_clus2  up/up      169.254.19.183/16  node2      e0b
true

4 entries were displayed.
```

19. 驗證兩個節點是否都與每個交換器有一個連線：「How cup neighbor」

    以下範例顯示兩個交換器的適當結果：

```
(cs1)# show cdp neighbors

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
ID
node1              Eth1/1         133    H           FAS2980       e0a
node2              Eth1/2         133    H           FAS2980       e0a
cs2                Eth1/35        175    R S I s     N9K-C9336C
Eth1/35
cs2                Eth1/36        175    R S I s     N9K-C9336C
Eth1/36

Total entries displayed: 4

(cs2)# show cdp neighbors

Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute

Device-ID          Local Intrfce  Hldtme Capability  Platform      Port
ID
node1              Eth1/1         133    H           FAS2980       e0b
node2              Eth1/2         133    H           FAS2980       e0b
cs1                Eth1/35        175    R S I s     N9K-C9336C
Eth1/35
cs1                Eth1/36        175    R S I s     N9K-C9336C
Eth1/36

Total entries displayed: 4
```

20. 顯示叢集中探索到的網路裝置相關資訊：「network device-dDiscovery show -protocol cup」

```
cluster1::*> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e0a    cs1                       0/2               N9K-
C9336C
            e0b    cs2                       0/2               N9K-
C9336C
node1      /cdp
            e0a    cs1                       0/1               N9K-
C9336C
            e0b    cs2                       0/1               N9K-
C9336C

4 entries were displayed.
```

21. 確認設定已停用：「network options switchless cluster show」

    命令可能需要幾分鐘的時間才能完成。等待「3分鐘壽命即將到期」公告。

    下列範例中的假輸出顯示組態設定已停用：

```
cluster1::*> network options switchless-cluster show
Enable Switchless Cluster: false
```

22. 驗證叢集中節點成員的狀態：「cluster show」（叢集顯示）

    下列範例顯示叢集中節點的健全狀況和資格資訊：

```
cluster1::*> cluster show

Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------  --------
node1                true    true          false
node2                true    true          false
```

23. 確保叢集網路具備完整連線能力：「cluster ping-cluster -node-name」（叢集ping叢集節點節點名稱）

```
cluster1::*> cluster ping-cluster -node node2
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1 e0a
Cluster node1_clus2 169.254.49.125 node1 e0b
Cluster node2_clus1 169.254.47.194 node2 e0a
Cluster node2_clus2 169.254.19.183 node2 e0b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:

Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)

Detected 9000 byte MTU on 4 path(s):
Local 169.254.47.194 to Remote 169.254.209.69
Local 169.254.47.194 to Remote 169.254.49.125
Local 169.254.19.183 to Remote 169.254.209.69
Local 169.254.19.183 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

24. 將權限等級變更回admin：「et -priv. admin」
25. 針對《支援乙太網路》9.8及更新版本、請使用下列命令啟用乙太網路交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔：「系統交換器乙太網路記錄設定密碼」及「系統交換器乙太網路記錄啟用-收集」ONTAP

```
cluster1::*> system switch ethernet log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system switch ethernet log setup-password

Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system switch ethernet log setup-password

Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system switch ethernet log enable-collection

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

26. 對於9.5P16、9.6P12和9.7P10及更新版本的修補程式、請使用「系統叢集交換器記錄設定密碼」和「系統叢集交換器記錄啟用收集」命令、啟用乙太網路交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔ONTAP

```
cluster1::*> system cluster-switch log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system cluster-switch log setup-password

Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y

Enter the password: <enter switch password>
Enter the password again: <enter switch password>

cluster1::*> system cluster-switch log enable-collection

Do you want to enable cluster log collection for all nodes in the
cluster?
{y|n}: [n] y

Enabling cluster switch log collection.

cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

27. 如果您禁止自動建立個案、請叫用AutoSupport 下列訊息重新啟用此功能：「System Node AutoSupport 現象叫用節點*-type all -most MAn=end」

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.